Chapter 3 Step 1: Problem Definition

The problem definition step in algorithmic trading focuses on the identification of **the specific financial objective the algorithm aims to achieve,** such as predicting stock prices, optimizing trade execution, or managing risk through dynamic portfolio adjustments.

The chosen financial objective is then translated into **the target variable you intend to predict or optimize**—for example, a stock's future price, the next period's market volatility, or the expected return on a portfolio.

Once the target variable is identified, the next step is to specify **the scope and constraints of the problem**, such as defining the time frame for predictions (e.g., intraday, daily, weekly), the markets or assets to be included (e.g., equities, commodities, forex), any regulatory or operational constraints that must be considered, as well as the trading strategy's risk tolerance and performance benchmarks.

A well-defined problem also includes understanding the relationships between the target variable, **known as labels or dependent variables,** and potential predictor variables, **known as features or fac-**

tors or independent variables. These features can include historical price data, trading volumes, economic indicators, and even sentiment analysis from news articles or social media. Establishing **a clear hypothesis about how these features interact with the target variable** will inform the data collection and feature engineering processes.

Let’s illustrate this step with three case studies.

CASE STUDY 1: FORECASTING SHORT-TERM STOCK MARKET TRENDS

In this case study, the **financial objective** is to identify profitable trading opportunities within a daily time frame, with **the target variable** being the stock's closing price for the next trading day.

The **time frame** of the problem is daily predictions, and our **focus assets and markets** are equities from major stock exchanges, such as the NYSE and NASDAQ.

The **regulatory constraints** are compliance with relevant financial regulations, while the **operational constraints** are transaction costs and potential liquidity issues.

The **potential features** include the following:

- Historical Price Data: historical closing prices and high, low, and open prices.
- Trading Volumes: daily trading volumes.
- Technical Indicators: Moving Averages (MA), Relative Strength Index (RSI), and Bollinger Bands.
- Economic Indicators: economic data like interest rates and inflation rates.
- Sentiment Analysis: sentiment scores from financial news articles and social media.

We **hypothesize** that patterns in historical price, trading volumes, technical indicators, economic indicators, and sentiment analysis can provide varying degrees of predictive insights into the stock's future closing price.

CASE STUDY 2: MITIGATING RISK WITH ADAPTIVE PORTFOLIO REBALANCING

The problem definition step in this case study sets the **financial objective** of maximizing risk-adjustment returns while maintaining a desired risk profile. The **target variable** for this problem is the portfolio's risk-adjusted return, such as the Sharpe ratio.

The **portfolio constraints** are asset allocation limits and diversification rules.

The **time frame** is weekly portfolio adjustments, and we limit our **assets and markets** to US equities, bonds, and commodities.

The **regulatory constraints** include compliance with portfolio management regulations, and the **operational constraints** are transaction costs and rebalancing frequency.

The **potential features** include the following:

- Market Data: historical prices and returns of portfolio assets.
- Risk Metrics: volatility, value-at-risk (VaR), and asset correlations.
- Economic Indicators: macroeconomic data-impacting asset classes.
- Sentiment Analysis: market sentiment from news articles and social media.

We **hypothesize** that actively altering portfolio composition in response to market dynamics and risk assessments could improve the portfolio's returns when factoring in risk while adhering to a specified level of risk exposure.

CASE STUDY 3: ENHANCING TRADE EXECUTION WITH REINFORCEMENT LEARNING TECHNIQUES

In this case study, our **financial objective** is to optimize trade execution to minimize transaction costs and market impact. The **target variable** for this problem is the execution price improvement compared to a benchmark price, such as the volume-weighted average price (VWAP).

The **time frame** is intraday trading; we focus on high-liquidity equities and exchange traded fund (ETF) **assets** in the US **markets.**

The **regulatory constraints** are compliance with market regulations and trading rules, and the **operational constraints** are order size, market impact, and liquidity.

The **potential features** include the following:

- Market Data: real-time order book data, trade prices, and volumes.
- Benchmarks: volume-weighted average price (VWAP) and time-weighted average price (TWAP).
- Order Execution Details: order size, type (market, limit), and execution time.
- Economic Indicators: relevant economic events and news that might impact market liquidity.
- Historical Execution Data: past execution performance and slippage data.

The **hypothesis** is that reinforcement learning algorithms applied to the trade execution process can reduce transaction costs and improve execution prices by better order placement and timing.
